Project 2 — Network Discovery & Vulnerability Assessment

Beginner–Intermediate | Suggested duration: 6–8 hours

Case study

VI-Systems has a small internal training network with three authorized lab VMs. The IT team suspects undocumented services and outdated software. Discover the lab assets, identify exposed services, review scanner results, and produce a defensible remediation plan.

Student challenge

Complete all seven tasks, retain evidence for each task, analyze the results, and submit a professional report with defensible conclusions and prioritized recommendations.

Lab environment and tools

Kali Linux; isolated host-only/internal virtual network; instructor-provided LAB_SUBNET and LAB_TARGETS; Nmap; optional Greenbone/OpenVAS. Never substitute a public subnet.

Tasks and required evidence

Task 1: Confirm scope and connectivity

What to do: Record the exact lab subnet and targets; confirm the scanning VM is attached to the approved isolated network.

Suggested tool(s): VirtualBox/VMware network settings; ip/ifconfig; ipconfig

Command / procedure:

```
Linux: ip addr; ip route. Windows: ipconfig /all. Compare the active interface and route with the instructor-provided lab range.
```

Evidence to submit: Screenshot of VM network mode, IP configuration, and approved target list.

What to explain: Explain the difference between the tester IP, target IP, and subnet.

Task 2: Discover approved lab hosts

What to do: Identify responsive systems only inside the supplied isolated lab subnet.

Suggested tool(s): Nmap

Command / procedure:

```
sudo nmap -sn LAB_SUBNET
```

Evidence to submit: Nmap output saved as evidence; list discovered hosts and how you confirmed they are in scope.

What to explain: Explain that host discovery can miss systems that block probes.

Task 3: Scan selected lab ports

What to do: Run a basic TCP port scan against each specifically approved target.

Suggested tool(s): Nmap

Command / procedure:

```
nmap -sT -T3 --top-ports 100 -oN task3_ports.txt LAB_TARGET
```

Evidence to submit: Screenshot plus task3_ports.txt. Replace LAB_TARGET with one approved lab IP.

What to explain: Explain open, closed, and filtered results; do not infer a vulnerability from an open port alone.

Task 4: Identify service versions

What to do: Collect service/version information from the approved target and compare it with the initial scan.

Suggested tool(s): Nmap

Command / procedure:

```
nmap -sV --version-light -T3 -oN task4_services.txt LAB_TARGET
```

Evidence to submit: Saved output, screenshot, and a table of port, service, detected version, and confidence.

What to explain: Explain version-detection limitations and why findings need validation.

Task 5: Review vulnerability scanner output

What to do: Run an approved authenticated or unauthenticated scan against the lab target, if the instructor has provided a scanner.

Suggested tool(s): Greenbone/OpenVAS or instructor-supplied report

Command / procedure:

```
Use the scanner UI: select only LAB_TARGET, choose a non-destructive assessment profile, start scan,
export report.
```

Evidence to submit: Exported scanner report and screenshot showing target scope and scan completion.

What to explain: Describe false positives, stale signatures, and the need to validate results.

Task 6: Validate and rank findings

What to do: Compare scanner results with service evidence and vendor advisories supplied by the instructor.

Suggested tool(s): Spreadsheet; Nmap results; scanner report

Command / procedure:

```
Create a table: finding, asset, evidence, scanner confidence, business impact, severity, validation
status.
```

Evidence to submit: Completed findings register with references to the exact evidence.

What to explain: Prioritize using impact and exposure, not just scanner severity labels.

Task 7: Deliver remediation plan

What to do: Recommend service removal, patching, configuration changes, segmentation, or monitoring as appropriate.

Suggested tool(s): Word processor / spreadsheet

Command / procedure:

```
For each finding, add action, owner role, urgency, validation test, and residual risk.
```

Evidence to submit: Final report, topology diagram, scan outputs, and prioritized remediation tracker.

What to explain: Explain which fixes should be done first and how the administrator will confirm closure.

Project completion checklist

- All seven tasks completed or any blockers clearly documented.
- Every task has a screenshot, exported output, or documented reason why evidence could not be obtained.
- Screenshots include readable tool context and a short caption; secrets and personal data are redacted.
- Findings are tied to evidence and severity is justified.
- The final report includes scope, methodology, results, limitations, remediation, and conclusion.